Quick Work

Page No.:

Date:

M T W T F S S

Experiment-1 Basic Commands.

Study of various network commands used in linux and windows commands.

1. arp -?

```
C:\Users\Admin>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a               Displays current ARP entries by interrogating the current
                   protocol data.  If inet_addr is specified, the IP and Physical
                   addresses for only the specified computer are displayed.  If
                   more than one network interface uses ARP, entries for each ARP
                   table are displayed.
  -g               Same as -a.
  -v               Displays current ARP entries in verbose mode.  All invalid
                   entries and entries on the loop-back interface will be shown.
  inet_addr        Specifies an internet address.
  -N if_addr       Displays the ARP entries for the network interface specified
                   by if_addr.
  -d               Deletes the host specified by inet_addr. inet_addr may be
                   wildcarded with * to delete all hosts.
  -s               Adds the host and associates the Internet address inet_addr
                   with the Physical address eth_addr.  The Physical address is
                   given as 6 hexadecimal bytes separated by hyphens. The entry
                   is permanent.
  eth_addr         Specifies a physical address.
  if_addr          If present, this specifies the Internet address of the
                   interface whose address translation table should be modified.
                   If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

2. hostname -?

```
C:\Users\Admin>hostname
RTS
```

3. netstat -?

```
C:\Users\Admin>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49679        RTS:49680              ESTABLISHED
  TCP    127.0.0.1:49680        RTS:49679              ESTABLISHED
  TCP    127.0.0.1:49681        RTS:49682              ESTABLISHED
  TCP    127.0.0.1:49682        RTS:49681              ESTABLISHED
```

4. ipconfig

```
C:\Users\Admin>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3616:d7ee:4b89:3108%16
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

5. nbtstat

```
C:\Users\Admin>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

  -a   (adapter status) Lists the remote machine's name table given its name
  -A   (Adapter status) Lists the remote machine's name table given its
                        IP address.
  -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addre
  -n   (names)          Lists local NetBIOS names.
  -r   (resolved)       Lists names resolved by broadcast and via WINS
  -R   (Reload)         Purges and reloads the remote cache name table
  -S   (Sessions)       Lists sessions table with the destination IP addresses
  -s   (sessions)       Lists sessions table converting destination IP
                        addresses to computer NETBIOS names.
  -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

  RemoteName   Remote host machine name.
  IP address   Dotted decimal representation of the IP address.
  interval     Redisplays selected statistics, pausing interval seconds
               between each display. Press Ctrl+C to stop redisplaying
               statistics.
```

6. nslookup

```
C:\Users\Admin>nslookup www.rajalakshmi.edu.in
Server:  UnKnown
Address:  10.99.125.84

Non-authoritative answer:
Name:    www.rajalakshmi.edu.in
Addresses:  64:ff9b::76b9:bb85
          64:ff9b::a2ff:77fd
          118.185.187.133
          162.255.119.253
```

7. pathping

```
C:\Users\Admin>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
  -g host-list     Loose source route along host-list.
  -h maximum_hops  Maximum number of hops to search for target.
  -i address       Use the specified source address.
  -n               Do not resolve addresses to hostnames.
  -p period        Wait period milliseconds between pings.
  -q num_queries   Number of queries per hop.
  -w timeout       Wait timeout milliseconds for each reply.
  -4               Force using IPv4.
  -6               Force using IPv6.
```

8. ping

```
C:\Users\Admin>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```